Closed loop control of three-phase AC-DC converter-I

Introduction

In the rapidly evolving landscape of electric vehicle charging and grid integration, efficient and reliable power conversion is paramount. Three-phase AC-DC converters, often found at the heart of these systems, play a critical role in converting alternating current from the grid into direct current suitable for battery charging or other DC loads. While understanding the basic operation and component sizing of these converters is essential, achieving optimal performance requires sophisticated control strategies. This lecture delves into the crucial aspect of closed-loop control for three-phase AC-DC converters, focusing on how to precisely regulate output voltage and ensure unity power factor operation, which are vital for grid stability and energy efficiency.

The challenge lies in managing the inherent interdependencies of three-phase systems. Unlike single-phase converters where dual-loop control (voltage and current) suffices, three-phase systems present a more complex control problem due to the coupled nature of their currents. This lecture introduces powerful mathematical transformations that allow us to decouple these quantities, enabling independent control and unlocking superior performance from our converters. By understanding these techniques, we can design systems that not only deliver stable power but also interact harmoniously with the electrical grid.

Core Concepts

Term	Definition	Significance
`Unity Power Factor (UPF)`	A condition where the current drawn from the source is perfectly in phase with the voltage, resulting in minimal reactive power exchange.	Essential for efficient power transfer, reducing grid losses, and minimizing harmonic distortion, ensuring the converter acts as a purely resistive load.
`Independent Control`	The ability to adjust one control variable without affecting other control	Crucial for precise and stable regulation of multiple converter parameters (e.g., active and reactive power, or

Term	Definition	Significance
	variables in a multi-variable system.	voltage and current components) simultaneously.
`Space Vector`	A single rotating vector that represents the combined instantaneous information of three-phase quantities (voltages or currents) in a two-dimensional plane.	Simplifies the analysis and control of three-phase systems by consolidating three interdependent quantities into one rotating vector, making it easier to visualize and manipulate.
`Stationary Reference Frame (Alpha-Beta)`	A two-axis coordinate system (alpha and beta axes) fixed in space, where the alpha axis is typically aligned with one of the phase axes (e.g., A-phase).	Provides a transformation from three-phase quantities to two orthogonal, independent components, simplifying control by eliminating the dependency inherent in the ABC frame.
`Rotating Reference Frame (DQ)`	A two-axis coordinate system (D and Q axes) that rotates synchronously with the space vector of the three-phase quantities. The D-axis is typically aligned with the space vector.	Further simplifies control by transforming AC quantities into DC quantities in the rotating frame, allowing for the use of conventional DC controllers (e.g., PI controllers) for precise regulation.
`Component Dependency`	The characteristic of three-phase quantities (like currents) where a change in one phase directly influences the others, as their sum must always be zero in a balanced system.	Highlights the fundamental challenge in directly controlling three-phase systems, necessitating transformations to achieve independent control over individual components.

Detailed Analysis

1. Converter Design Recap and Control Objectives

Before diving into control, it's important to recall the foundational design aspects of the three-phase AC-DC converter. Previous discussions covered the sizing of critical components:
* **Inductance Value**: Calculated to ensure the maximum voltage drop across the inductor, corresponding to the fundamental frequency component, remains below 10% of the maximum input voltage. Exceeding this limit would lead to excessive voltage drop across the inductor itself, hindering converter operation.
* **Capacitance Value**: Determined based on the desired output voltage ripple (ΔV0), the maximum power variation (ΔPmax), and the closed-loop response time (TD). The RMS current flowing through the capacitor bank, which is independent of switching frequency and depends on the modulation index (M) and maximum source current (ISP), is also crucial for designing the series-parallel combination of capacitors. The voltage rating of the output capacitance must exceed V0 + ΔV0/2.
* **Switch Ratings**: RMS current calculations for the switches and their voltage blocking capabilities are essential for selecting appropriate power semiconductor devices.

With the converter components designed, the primary control objectives for its operation are:
* **Output Voltage Regulation**: Maintain the DC output voltage (V0) at a desired, constant value. This value must be greater than the peak line-to-line voltage (√2 * Vline-to-line), typically set around 650-700V to account for drops and provide sufficient headroom.
* **Unity Power Factor (UPF) Operation**: Ensure that the current drawn from the three-phase AC source is in phase with the respective phase voltages (e.g., ISA in phase with VAN, ISB with VBN, ISC with VCN). This minimizes reactive power exchange and improves overall system efficiency.

2. The Challenge of Three-Phase Current Control

In a balanced three-phase system, the sum of the instantaneous phase currents (ISA + ISB + ISC) is always zero. This fundamental property implies that the three phase currents are not independent of each other. If one attempts to control the currents in phase A and phase B, the current in phase C is automatically determined. This dependency makes direct, independent control of all three phase currents challenging. For instance, if we try to control phase A current, changes will inevitably occur in phases B and C, making it difficult to achieve precise UPF operation across all phases simultaneously using conventional single-phase control techniques. To

overcome this, a transformation from the three-phase system to a two-phase system is necessary to achieve independent control.

3. The Space Vector Concept

To simplify the representation and control of three-phase quantities, the concept of a "space vector" is introduced. Imagine an observer moving synchronously with the time-varying three-phase voltages (VAN, VBN, VCN). These voltages are 120 degrees apart in time. If we represent these voltages as spatial vectors also 120 degrees apart (A-axis, B-axis, C-axis), the resultant of these three vectors forms a single "space vector."

For example, if the observer aligns with the peak of the A-phase voltage (at ωt = 90 degrees), the A-phase contributes its peak value (VSP), while the B and C phases contribute -VSP/2 each. The resultant vector, when calculated by summing the components along the A-axis, is 3/2 VSP. Similarly, if the observer moves to align with the peak of the B-phase or C-phase, the resultant magnitude remains 3/2 VSP. This demonstrates that the combined effect of the three-phase quantities can be represented by a single vector whose magnitude is constant (3/2 VSP) and which rotates in a two-dimensional space at the angular speed 'ω' (corresponding to the line frequency). This space vector encapsulates all the information of the three-phase quantities.

4. Stationary Reference Frame (Alpha-Beta Transformation)

The space vector rotates in a two-dimensional plane. To achieve independent control, we need to represent this rotating vector using two orthogonal axes. The stationary reference frame, also known as the alpha-beta frame, provides this.
* **Definition**: The alpha-beta frame consists of two axes, alpha (α) and beta (β), fixed in space and oriented 90 degrees apart. The alpha axis is typically aligned with the A-phase axis of the three-phase system.
* **Independence**: The key advantage of this frame is that the projection of any vector onto the alpha axis is completely independent of its projection onto the beta axis, and vice-versa. This is because the cosine of 90 degrees is zero, meaning there is no cross-coupling between the components.
* **Transformation**: Three-phase quantities (VA, VB, VC) can be transformed into their alpha-beta components (Vα, Vβ) using trigonometric calculations. For instance, Vα would primarily depend on VA, while Vβ would depend on a combination of VB and VC, ensuring their orthogonality.
* **Control Benefit**: By controlling Vα and Vβ independently, we can indirectly control the three-phase quantities. Any change in Vα will not affect Vβ, allowing for decoupled control.

5. Rotating Reference Frame (DQ Transformation)

While the alpha-beta frame provides independent control, the components (Vα, Vβ) are still AC quantities, varying sinusoidally as the space vector rotates. To further simplify control, especially for using conventional DC controllers (like PI controllers), the rotating reference frame, or DQ frame, is employed.
* **Definition**: The DQ frame consists of two axes, direct (D) and quadrature (Q), which rotate synchronously with the space vector at the angular speed 'ω'. The D-axis is typically aligned with the instantaneous position of the space vector, and the Q-axis is 90 degrees ahead of it.
* **Synchronous Rotation**: Because the DQ frame rotates with the space vector, the components of the space vector when projected onto the D and Q axes appear as constant DC quantities (VD, VQ) in the steady state.
* **Independence**: Similar to the alpha-beta frame, the D and Q axes are 90 degrees apart, ensuring that VD and VQ are completely independent of each other.
* **Transformation**: The alpha-beta components (Vα, Vβ) can be transformed into DQ components (VD, VQ) using a rotational transformation that accounts for the angle (θ = ωt) of the rotating frame.
* **Control Benefit**: By transforming AC quantities into DC quantities in the DQ frame, standard PI controllers, which are highly effective for DC regulation, can be used to control VD and VQ independently. This allows for precise control over active and reactive power components, which are directly related to the D and Q axis currents, respectively. For example, controlling the D-axis current allows control over active power, while controlling the Q-axis current allows control over reactive power, enabling UPF operation.

These transformations (ABC to alpha-beta, and alpha-beta to DQ) are fundamental tools in the closed-loop control of three-phase AC-DC converters, enabling the independent manipulation of system variables that are otherwise coupled.

Key Takeaways

- [ ] Three-phase AC-DC converters require closed-loop control to regulate output voltage and achieve unity power factor.
- [ ] Component sizing (inductance, capacitance, switch ratings) is a prerequisite for effective control system design.
- [ ] The sum of instantaneous three-phase currents is zero, making direct independent control of each phase current impossible.
- [ ] The space vector concept represents three-phase quantities as a single rotating vector, simplifying analysis.

- [ ] The stationary alpha-beta reference frame transforms three-phase quantities into two orthogonal, independent AC components.
- [ ] The rotating DQ reference frame transforms alpha-beta components into two orthogonal, independent DC components.
- [ ] DQ transformation enables the use of standard DC controllers (e.g., PI) for precise control of active and reactive power.
- [ ] Independent control in the DQ frame is crucial for achieving both output voltage regulation and unity power factor operation.

Conclusion

The journey from understanding the basic operation of a three-phase AC-DC converter to implementing sophisticated closed-loop control is paved by the mastery of reference frame transformations. The inherent dependency of three-phase quantities poses a significant challenge to precise control. However, by leveraging the space vector concept and transforming these quantities into orthogonal, independent components within stationary (alpha-beta) and rotating (DQ) reference frames, we unlock the ability to achieve decoupled control.

This foundational understanding is critical for designing high-performance converters that can maintain stable DC output voltage while ensuring efficient, unity power factor interaction with the AC grid. The ability to independently control active and reactive power components in the DQ frame is not just an academic exercise; it is a practical necessity for modern power electronics applications, from electric vehicle charging stations to renewable energy integration. This lecture sets the stage for further exploration into the detailed implementation of these control strategies, paving the way for robust and intelligent power conversion systems.